Rajdeep Gill

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Education

University of Manitoba May 2026
B.S. Computer Engineering **4.46/4.5 GPA**
Coursework: Data Structures and Algorithms, Parallel Computing, Digital Signal Processing

Experience

Robinhood **May 2025 - Sept 2025**
Software Engineer Intern *Toronto, ON*

- Working on creating a dashboard to visualize and handle margin calls and liquidations for Robinhood's brokerage team.
- Implementing integration tests to ensure the reliability of the dashboard, and working on improving the performance of the dashboard by optimizing the data fetching and processing.

Biosensors Research Lab **May 2024 - Dec 2024**
Undergraduate Research Student *Winnipeg, MB*

- Developed and trained a CNN to predict health of cells based on scattering images, achieving 85% accuracy.
- Created a data visualization tool to display cell health data based on the model's predictions.
- Improved parameter extraction by 17% with a custom centroid detection algorithm, making use of dynamic thresholding and edge detection.
- Developed a data extraction pipeline with OpenCV and parallelizing with OpenMP, reducing processing time by 40%.
- Built a SQL database to store experimental data and processed results, along with a REST API to query the data.

Biosensors Research Lab **May 2023 - Aug 2023**
Undergraduate Research Student *Winnipeg, MB*

- Developed an automated oscilliscope testing platform with PyVISA to streamline data collection, reducing manual collection time and ensuring repeatability.
- Optimized existing data analysis algorithms with vectorization and parallelization, reducing processing time by 90%.
- Explored the use of unsupervised learning to cluster data points based on their impedance spectra.
- Implemented a prediction model based on a Gaussian Mixture Model and probabilistic neural networks to predict the health of cells based on their impedance spectra.

Projects

Distributed Image Processing *(C++, OpenMPI)*

- Created a distributed image processing tool to process large images across multiple nodes, achieving a 95% efficiency.
- Implemented a custom communication protocol to optimize data transfer latency and throughput.
- Utilized OpenMPI to handle the distribution of tasks across multiple nodes, ensuring scalability and reliability.

Collaborative Document Editor *(React, TypeScript, Socket.io, Flask)*

- Built a collaborative document editor enabling real-time multi-user editing and communication.
- Implemented bidirectional messaging with Socket.io for seamless updates, and developed a RESTful API using Flask.
- Enhanced user experience by integrating autocomplete and in-line suggestions powered by Google's Gemini API.

Financial Dashboard *(React, TypeScript, Hono, PostgreSQL)*

- Developed a financial dashboard to visualize financial data, allowing users to track their investments and expenses.
- Utilized Hono to create a REST API to fetch and store financial data, and PostgreSQL to store the data.
- Implemented React to create a user-friendly interface, allowing users to visualize their financial data.

Interpretor *(C++)*

- Developed an interpretor for a custom language to allow users to write and run code in a terminal.
- Implemented a lexer and parser to tokenize and parse the code, and a semantic analyzer to check for errors.

Network Simulator *(Java)*

- Developed a network simulator with packet loss, congestion and corruption to test various protocols.
- Implemented Alternate Bit, Go-Back N, Selective Repeat and TCP to test under various conditions.

Skills

Programming Languages: C++, Python, TypeScript, JavaScript, Java
Frameworks and Libraries: React, NextJS, PyTorch, OpenMPI
Tools: Git, Docker, PostgreSQL
Concepts: Backend/Frontend Development, Machine Learning, Networking, CI/CD, Microservices, Agile